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1. Rumus Naive Bayes

$F(x) = \text{argmax}_y \, P(Y=y) \times P(X=x \mid Y=y)$

- $P(Y=y)$ = prior probability
- $P(X \mid Y) = \prod_{i=1}^{n} p(x_i \mid y)$

2. Data Statistik

kelas 0 (tidak diabetes) = 401

kelas 1 (diabetes) = 213 data

Total data = 613

3. Menghitung Prior Probability

- $P(Y=0) = \frac{401}{613} = 0,6531$
- $P(Y=1) = \frac{213}{613} = 0,3469$

4. Menghitung ~~Prior Probability~~ Likelihood

$P(x_i \mid Y) = \frac{\text{count}(x_i, y)}{\text{count } y}$

Fitur 1 = Pregnancies:

kelas 0

$P(x_1 \mid 0) = \frac{320}{401} = 0,7980$

kelas 1

$P(x_1 \mid 1) = \frac{50}{213} = 0,2347$

Fitur 2 = Glucose

$P(x_2 \mid 0) = \frac{300}{401} = 0,7481$

kelas 1

$P(x_2 \mid 1) = \frac{40}{213} = 0,1878$

Fitur 3 = Blood pressure

$P(x_3 \mid 0) = \frac{350}{401} = 0,8728$

kelas 1

$P(x_3 \mid 1) = \frac{100}{213} = 0,4695$

Fitur 4 = Skin thickness

$P(x_4 \mid 0) \frac{280}{401} = 0,6983$ (kelas 0)

$P(x_4 \mid 1) = \frac{90}{213} = 0,4225$ (kelas 1)

Fitur 5 = Insulin

kelas 0

$P(x_5 \mid 0) = \frac{310}{401} = 0,7731$

kelas 1

$P(x_5 \mid 1) = \frac{80}{213} = 0,3756$

Fitur 6 = BMI

kelas 0

$P(x_6 \mid 0) = \frac{290}{401} = 0,7232$

kelas 1

$P(x_6 \mid 1) = \frac{60}{213} = 0,2817$

Fitur 7 = Diabetes Pedigree

kelas 0

$P(x_7 \mid 0) = \frac{340}{401} = 0,8479$

kelas 1

$P(x_7 \mid 1) = \frac{75}{213} = 0,3521$

Fitur 8 = Age

kelas 0

$P(x_8 \mid 0) = \frac{360}{401} = 0,8978$

kelas 1

$P(x_8 \mid 1) = \frac{45}{213} = 0,2113$

5. Menghitung Posterior probability

kelas 0 (tidak diabetes):

$P(Y=0) \times P(x_1 \mid 0) \times P(x_2 \mid 0) \times P(x_3 \mid 0) \times P(x_4 \mid 0) \times P(x_5 \mid 0) \times P(x_6 \mid 0) \times P(x_7 \mid 0) \times P(x_8 \mid 0)$

$= 0,6531 \times 0,7980 \times 0,7481, 0,8728 \times 0,6983 \times 0,7731 \times 0,7232 \times 0,8479 \times 0,8978$

$= 0{,}0925$

kelas 1 (diabetes)

$= P(Y=1) \times P(x_1|1) \times P(x_2|1) \times P(x_3|1) \times P(x_4|1) \times P(x_5|1) \times P(x_6|1) \times P(x_7|1) \times P(x_8|1)$

$= 0{,}3469 \times 0{,}2347 \times 0{,}1878 \times 0{,}4695 \times 0{,}4225 \times 0{,}3756 \times 0{,}2817 \times 0{,}3521 \times 0{,}2113$

$= 0{,}0000236$

6. kesimpulan

karena nilai score kelas 0 (0,0925) > score kelas 1 (0,0000236), maka berdasarkan perhitungan naive Bayes, data tersebut diklasifikasikan sebagai kelas 0 (tidak diabetes).